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INDUSTRIAL ENGINEERING

Interrelationship model of sustainability indicators for the agro-industrial supply chain in the Andean region

- Mora Asanza Oscar Antony
- Romero Bravo Alison Camila
- **Tutor:** Ing. Pablo Andrés Flores Siguenza

Abstract

The Andean agro-industry faces sustainability challenges that require integrated approaches. This study develops a model of interrelationships between indicators to analyze the sustainable performance of an agro-industrial chain from a systemic perspective.

Objectives

General Objective

Develop a model for the integration and interrelation of sustainability indicators to improve operational efficiency and strategic decision-making in the agro-industrial supply chain in the Andean region.

Specific Objective 1 and 2

- Design and implement a tool to filter and select, from an existing framework, the most relevant sustainability indicators for an Andean agro-industrial organization.
- Measure the selected indicators in a case study agro-industrial organization to generate contextualized information.

Specific Objective 3

Modeling the interdependencies between environmental, economic and social indicators, identifying significant causal relationships and their sensitivity to different scenarios proposed in the case study.

Specific Objective 4

- To propose a digital tool that allows visualizing the interrelationship between indicators for strategic decision-making.

Methodology

1-2
- Phase I. Grouping of indicator packages
- Phase II. Design and application of the selection tool

3-4
- Phase III. Prioritization of sustainability indicators
- Phase IV. Measurement of indicators

5-6
- Phase V. Development of interrelationships between indicators (MICMAC).
- Phase VI. Visualization and application of scenarios

Curriculum Integration

- Production Scheduling: Applied mathematical model
- Analytical Statistics: Analysis and interpretation of data
- Strategic Information Systems: Interactive visualization

Applied Engineering

Standards: ISO 26000 and ISO 14031 conceptual references for the comprehensive understanding of sustainability and the use of indicators.
Design constraints of an environmental, economic and technological nature.

Keywords

Interrelationship model · Sustainability indicators · Agro-industry · MICMAC · Sustainability

Results

- Phase I - II: 135 indicators were reorganized into Basic, Intermediate and Advanced levels, identifying the Basic Level as the most appropriate for the case study, as it represents the minimum sustainability capabilities.

- Phase III: From the basic package, 21 indicators were prioritized, aligned to the real context of the organization.
- Phase IV: The evaluation showed critical gaps in training (S-S-03, S-S-041), production (S-E-07) and income (S-E-020), and better performance in environmental practices and access to basic services.

- Phase V: The MICMAC analysis identified S-E-020 and S-S-03 as influential indicators, with relay indicators that articulate interdependencies in the short and long term.
- Phase VI: The scenarios confirmed that intervening S-E-020 generates the most favorable systemic impacts.

Conclusions

Conclusions 1

The developed model demonstrates that agroindustrial sustainability can be analyzed systemically. The progressive integration of selection, measurement, structural analysis and visualization allowed us to understand sustainable performance considering the interrelationships between indicators, strengthening strategic decision making in Andean agroindustrial contexts.

Conclusions 2

Contextual prioritization of indicators is key to ensuring applicability of the analysis. Based on a previously validated framework, the study managed to define a set of indicators aligned with the real functioning of the organization, ensuring relevance, coherence and practical usefulness to evaluate sustainability.

Conclusions 3

The MICMAC analysis showed a complex structure of influences and dependencies, where few indicators concentrate the capacity to generate systemic changes, while others reflect cumulative effects. Its relevance is manifested in the short term through direct relationships and in the long term through indirect relationships, conditioning the global behavior of the system.

Conclusions 4

Incorporating an interactive digital tool in Python significantly enhances the analysis and interpretation of complex systems. The integrated visualization and simulation of what-if scenarios reduce the cognitive load and facilitate the evaluation of alternatives, giving the model a high potential for replicability in other similar agroindustrial chains.

Contacts

Mora Asanza Oscar Antony
antony.mora99@ucuenca.edu.ec

Romero Bravo Alison Camila
camila.romerob@ucuenca.edu.ec

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